Day 19 — April 1, 2026 (Week 6, Day 2)

Real Traffic Testing, Attack Simulation & Demo Preparation

📋 PLANNED TASKS

#	Task	Priority	Status
1	Attack Simulation Scripts (nmap, hping3, hydra)	🔴	☐ PENDING
2	PCAP Demo Scenario Files (3-5 attack types)	🔴	☐ PENDING
3	E2E Real Traffic Walkthrough	🔴	☐ PENDING
4	LLM Narrative Quality Verification	🟡	☐ PENDING
5	Auth Enable + Demo Account Creation	🟡	☐ PENDING
6	VPS System Health Verification	🔴	☐ PENDING

Context & Rationale

Day 18 completed the full frontend overhaul with all 10 pages connected to the VPS backend. A comprehensive project audit (9 dimensions) identified the **#1 gap**: no attack simulation scripts or PCAP demo scenarios exist for demo day presentations. Per MASTER_DOC_PART5 §7.3 (Demo Scenario Test Cases) and §8.1 (Demo Script), the system must demonstrate live anomaly detection with visible alert generation.

v0.6.0 is achieved — 1 week ahead of schedule. This buffer should be used for demo readiness, which is the critical path to a successful presentation.

Task 1: Attack Simulation Scripts 🔴

Reference: PART5 §7.3 Demo Scenario Test Cases

Create executable scripts for the following attack scenarios against the VPS:

Scenario	Tool	Expected Detection	Alert Severity
Port Scan	`nmap -sS target_ip`	Probe detected, port sweep classification	HIGH
DDoS Simulation	`hping3 --flood target_ip`	Volume anomaly, DDoS classification	CRITICAL
DNS Tunneling	Crafted DNS queries	Unusual DNS pattern, high entropy	MEDIUM

Scenario	Tool	Expected Detection	Alert Severity
Brute Force SSH	`hydra -l root -P wordlist ssh://target`	Failed login volume spike	HIGH
Normal Traffic	Regular browsing, API calls	No false alerts	—

Deliverable: `scripts/attack_simulation/` directory with documented scripts and expected outcomes.

Verification:

- ☐ Each attack produces ≥1 alert in the alerts table
- ☐ Alerts have correct severity classification
- ☐ LLM auto-narrative fires for each alert
- ☐ WebSocket broadcasts alert to connected frontend

Task 2: PCAP Demo Scenario Files

Reference: PART5 §8.2 Backup Plans — "Pre-loaded PCAP with interesting anomalies"

Create or obtain PCAP files representing distinct attack patterns for offline demo backup:

PCAP File	Content	Purpose
`ddos_scenario.pcap`	SYN flood / volumetric attack	Show CRITICAL alert generation
`port_scan.pcap`	nmap scan results	Show probe detection
`dns_tunnel.pcap`	DNS exfiltration pattern	Show entropy-based detection
`brute_force.pcap`	Repeated SSH/FTP attempts	Show R2L classification
`normal_traffic.pcap`	Clean traffic baseline	Show false-positive rate

Deliverable: `pcaps/demo/` directory with 3-5 PCAP files + README explaining each scenario.

Verification:

- ☐ Each PCAP uploads successfully via POST /capture/upload-pcap
- ☐ PCAP processor extracts flows and scores them
- ☐ Attack PCAPs produce alerts; normal PCAP produces none

Task 3: E2E Real Traffic Walkthrough

Reference: PART5 §8.1 Demo Script (5:30-10:30 mark)

Execute the demo script flow end-to-end:

1. Open War Room — verify live data flowing
2. Run nmap attack → verify alert fires within 60 seconds

3. Open Alert Console → verify alert detail with AI narrative
4. Open AI Analyst → ask about the attack → verify coherent response
5. Open ML Ops → verify model metrics display correctly
6. Open Reports → generate threat summary → download PDF
7. Open Intel Hub → verify IOC correlation data
8. Open Admin → verify audit log shows events

Deliverable: Written walkthrough results with pass/fail for each step.

Verification:

- ☐ Alert appears in War Room live feed within 60s of attack
- ☐ AI Analyst can explain the detected threat
- ☐ PDF report generates with correct data
- ☐ Total E2E latency (attack → alert visible) < 200ms target (PART4 §8.2)

Task 4: LLM Narrative Quality Check

Review the AI-generated narratives for recent alerts and verify:

- ☐ Narratives explain what happened clearly
- ☐ Why the activity is dangerous is stated
- ☐ Recommended actions are provided
- ☐ No hallucinations or incorrect technical details
- ☐ Language is professional and suitable for demo

If quality is poor, adjust prompt templates (PART4 §9.2) or switch to a better OpenRouter model.

Task 5: Auth Enable + Demo Accounts

Reference: PART5 §8.3 Pre-Demo Checklist — "Demo user account created (analyst role)"

1. Set `DEV_MODE=false` temporarily on VPS
2. Create demo accounts via auth endpoints:
   - `admin@threatmatrix.ai` (admin role)
   - `analyst@threatmatrix.ai` (analyst role)
   - `viewer@threatmatrix.ai` (viewer role)
3. Test login flow in frontend with each account
4. Verify RBAC enforcement (viewer can't retrain, analyst can analyze but not admin)

Note: Can revert to `DEV_MODE=true` after verification for continued development.

Task 6: VPS System Health Verification

Full infrastructure check:

Check	Command	Expected

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All containers running	`docker-compose ps`	5/5 up
Postgres health	`pg_isready -U threatmatrix`	OK
Redis health	`redis-cli ping`	PONG
Disk space	`df -h`	< 80% used
Memory usage	`free -h`	< 80% used
ML models loaded	`curl /api/v1/ml/models`	3 models
Capture active	`curl /api/v1/capture/status`	Running
Flow count growing	`curl /api/v1/flows/stats`	Count > 105,000
LLM responding	`curl /api/v1/llm/budget`	200 OK

Success Criteria for Day 19

- ☐ ≥3 attack types produce visible alerts in the frontend
- ☐ LLM narratives are generated and coherent for each alert
- ☐ At least 3 PCAP demo files created and verified
- ☐ E2E walkthrough documented with pass/fail
- ☐ Demo accounts created (can defer to later if auth flow has issues)
- ☐ VPS infrastructure verified healthy

Architectural Compliance Notes

All Day 19 tasks are **within the PART5 §7-8 scope** (Testing Strategy + Demo Day Preparation). No new modules, features, or technologies are being introduced. This is pure verification and demo readiness work.

Day 19 — PLANNED Focus: Real traffic testing → Attack simulation → Demo preparation Version: v0.6.0 (1 week ahead of schedule) Reference: PART5 §7.3 (Demo Scenarios), §8.1 (Demo Script), §8.3 (Pre-Demo Checklist)